



Quantitative modeling of near-field interactions incorporating polaritonic and electrostatic effects

G. CONRAD,^{1,2} C. B. CASPER,^{1,2}  E. T. RITCHIE,¹  AND J. M. ATKIN^{1,*} 

¹Department of Chemistry, University of North Carolina, Chapel Hill, NC, USA

²These authors contributed equally to this work

*jatkin@email.unc.edu

Abstract: As scattering-scanning near-field optical microscopy (s-SNOM) continues to grow in prominence, there has been great interest in modeling the near-field light-matter interaction to better predict experimental results. Both analytical and numerical models have been developed to describe the near-field response, but thus far models have not incorporated the full range of phenomena accessible. Here, we present a finite element model (FEM), capable of incorporating the complex physical and spatial phenomena that s-SNOM has proved able to probe. First, we use electromagnetic FEM to simulate the multipolar response of the tip and illustrate the impact of strong coupling on signal demodulation. We then leverage the multiphysics advantage of FEM to study the electrostatic effect of metallic tips on semiconductors, finding that THz s-SNOM studies are most impacted by this tip-induced band-bending. Our model is computationally inexpensive and can be tailored to specific nanostructured systems and geometries of interest.

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1. Introduction

Scattering-scanning near-field optical microscopy (s-SNOM) is a powerful tool for investigating material properties on deeply sub-diffraction length scales. In s-SNOM, an atomic force microscope (AFM) tip acts as an optical antenna, [1–3] confining incident light to a nanometer volume and scattering light into the far-field. [4] This tip-scattered field is only a small fraction of the overall back-scattered light but can be retrieved through lock-in filtering based on the tip's tapping motion, yielding an isolated near-field signal and images with spatial resolution on the order of the tip diameter. In its early stages, s-SNOM experiments were conducted for relatively simple scenarios such as gold structures on silicon, [5,6] planar semiconductors, [7,8] or polymer blends. [9] More recently, s-SNOM has been integrated with other broadband spectroscopies for the identification of vibrational modes in molecular [10–13] and biomolecular [14–16] systems as well as ultrafast experiments. [17–20] Solid state phenomena such as polaritonic properties [21–27] and the electrical characterization of semiconducting nanostructures and devices, [28–30] have also been explored using s-SNOM. In general, the refinement of the technique and emergence of commercialized instruments has led to an increase in s-SNOM experiments, many of which vary widely in their sample geometry, type of spectroscopy, and goals. [31]

With the application of s-SNOM to more complex samples there come commensurate challenges in analysis and interpretation. Even in simple scenarios, the interpretation of results in s-SNOM is complicated due to the mediating role of the tip in the near-field interaction, generating a coupled tip-sample response rather than a pure material response. [32] The most common model for this interaction, the tip-dipole model (TDM), approximates the AFM tip as a perfectly conducting spheroid polarized by a quasistatic electric field. The coupling of this spheroid with its image in the sample then leads to an effective polarizability that is used to estimate the complex near-field response. [1,33,34] The simplest case of the TDM is the point-dipole model (PDM), where the

tip is modeled as a sphere [5,33,35]. The PDM generally can predict simple material contrasts and provides good agreement for the qualitative behavior of weak molecular oscillators, but does not accurately predict spectral effects associated with stronger resonances. [36,37]. The model can be improved by treating the tip as an ellipsoid, as in the finite-dipole model (FDM), which reflects a more realistic elongated tip geometry [34] and can be further extended for the analysis of multilayered samples. [38,39] However, while these models are computationally straight-forward and allow direct comparison with experiment, they do not account for complex sample geometries, spatially-inhomogeneous samples, or strong tip-sample coupling responses.

Numerical techniques such as finite-element modeling (FEM) are an alternative method to model the coupled tip-sample response. While more computationally-intensive than the TDM, FEM is not limited by any geometric assumptions and can thus simulate realistic tip and sample geometries. In addition, FEM provides electric field maps that allow for the visualization of the near-field and its direct impact on the observed scattering. This is particularly significant for polaritonic materials, where the large k -vector of the nanoscale tip apex can launch polariton surface waves and produce complex optical patterns. [7,9,26,40–43] Thus, early FEM simulations investigated the field-enhancement and polaritonic properties of the strong phonon response in SiC. [22,32,44,45] More recent FEM simulations have aimed to establish a more direct link between models and s-SNOM observables. Mooshammer et al. developed a FEM model which provides the spatial distribution of the demodulated s-SNOM intensity, enabling new insight into the spatial profile of near-fields and their penetration depth for tomography in dielectrics. [46] A recent study by McArdle et al. utilized high-resolution electron microscopy and curve modeling to build a highly-realistic tip geometry and accurately predict the spectrum of a phonon-polariton material. [47] Spectra predicted by FEM models have recently been used to describe the role of tip geometry on the temporal response of ultrafast experiments on semiconductors in the THz regime. [48] These studies represent a significant advancement in modeling tip-sample interactions, but questions remain on implementing and utilizing FEM for the quantitative analysis of s-SNOM experiments.

Here, we demonstrate the ability of FEM to simulate complex systems encountered in s-SNOM experiments. We first model the demodulated amplitude and phase spectra for a relatively thin layer of polymethyl methacrylate (PMMA) and show the influence of the substrate on the simulated spectra. We then consider the strong-coupling case, investigating the multipolar near-fields associated with the SiC phonon-polariton resonance and their role in the demodulation process. Finally, we leverage the flexibility of FEM programs to incorporate multiple physical phenomena to model the electrostatic effects of metal s-SNOM tips on semiconductors, an effect not generally considered in near-field studies in these samples. We find a significant shift in the near-field response, especially at the lower terahertz (THz) frequencies, and demonstrate how this could impact the interpretation of measurements across semiconductor junctions. Our models can be run on a standard desktop computer and illustrate the wide applicability of FEM and its ability to provide detailed information of near-fields in a relatively short time.

2. Methods

FEM simulations are performed with the Wave Optics and Semiconductor modules in the commercial software COMSOL Multiphysics (v. 5.5 & 5.6). In general, we restrict our models to two dimensions (x, y) to reduce simulation time, which enables a single approach curve to be calculated in two hours with a standard desktop computer (Intel i7, 1.80 GHz, 16.0 GB). In Figure S2, we demonstrate that the dimensional reduction causes minimal loss of information, as the incident electromagnetic radiation is completely p-polarized, and the z -component (tangential) of the electric field has only a weak near-field dependence.

2.1. Scattering simulations

Here we model the tip as a platinum ellipse with major semi axis length $L = 300$ nm and apex radius $R = 35$ nm at a distance H above the surface of a sample with permittivity $\epsilon(\omega)$ (Fig. 1(A)). We choose an ellipse for ease of comparison to the FDM, but more realistic tip geometries, such as a parabola, can also be implemented with little effect on results (see Figure S2). For simulations in the infrared spectral range, we model our physical domain as a $30\ \mu\text{m} \times 30\ \mu\text{m}$ square region surrounded on all sides by a $20\ \mu\text{m}$ thick perfectly-matched layer. We scale this design up accordingly for simulations with longer wavelengths.

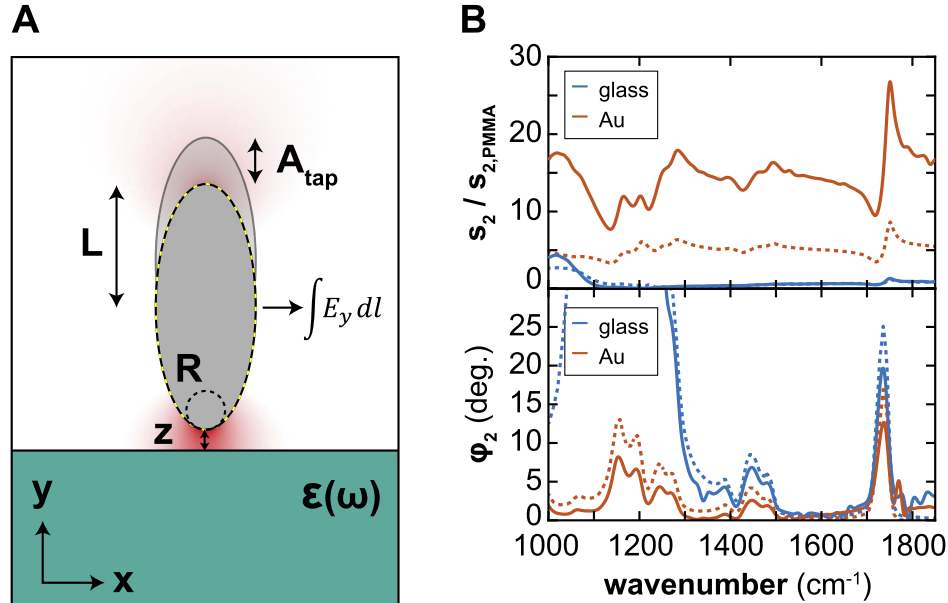


Fig. 1. (A) Schematic showing the finite-element approach for calculating the scattered near-field of a tip over a sample with permittivity $\epsilon(\omega)$. The tip is modeled as an ellipse with apex radius R and length L . We repeat the simulation at multiple heights to simulate the tapping mode A_{tap} necessary for demodulation. (B) Simulated near-field (2Ω) amplitude (top panel) and phase (bottom panel) for 10 nm of PMMA on glass (blue) and gold (orange) substrates. Dashed lines represent calculated spectra from the ML-FDM.

The scattering problem is solved in a two-step approach. In the first step, the top boundary of the model acts as a port and launches a plane wave towards the surface at an angle of 60° relative to the vertical. The resultant field is a superposition of the incident and reflected fields in the absence of any scatterers and is used as the input for the second step, which calculates the scattered field in the presence of the metallic tip. Previous work has used a number of different approaches for estimating the near-field signal from a given scattering model, such as point-evaluations underneath the AFM tip [47] and integral methods over the tip boundary [49] or a fictitious surface [46]. Here we choose to estimate the near-field signal by integrating the vertical component of the electric field over the surface of the tip. Unlike other methods, a tip integral remains accurate for cases with non-symmetric field distribution and avoids having to introduce any new parameters. Furthermore, the result of the integration is complex-valued, which can be converted into a near-field amplitude and phase for direct comparison with experimental results.

s-SNOM experiments are typically performed in tapping mode, with the tip oscillating above the sample with an amplitude on the order of tens of nanometers and frequency $\Omega \approx 50$ -300 kHz. [40,50] Because the tip-sample interaction is dependent on the distance from the surface,

the tapping motion modulates the near-field signal. By demodulating the near-field signal at harmonics of the tapping frequency (2Ω , 3Ω , etc.) then, the far-field background and scattering from other sources can be suppressed. Thus in order to compare directly to s-SNOM experiments, the FEM simulations must also be demodulated in a similar manner. We achieve this by performing simulations at multiple tip-sample distances with a 1 nm step size, simulating an approach curve that can be mirrored in time and then demodulated using a fast Fourier transform. [47] This results in simulated harmonics whose magnitude and argument can be directly compared with the experimentally measured near-field amplitude and phase. Tapping amplitudes stated in this work represent the mean distance between the tip and the sample with a minimum tip-sample separation of 1 nm.

For our scattering simulations in the IR, we use experimentally-derived permittivity data for: Pt, Au, PMMA, glass, and SiC. [36,51–54] For our THz simulations, we calculate a Pt permittivity using the Drude model with experimentally derived parameters. [55]

2.2. Electrostatic simulations

Electrostatic effects on the near-field signal in semiconductors are simulated by first using the Semiconductor module to solve the drift and diffusion equations for a metallic tip some distance above doped Si. The tip is not modeled explicitly, but instead is replaced with a Terminal boundary condition with Pt work function $\phi = 5.4$ V and applied voltage $V_a = 0$ V. Carriers within the Si domains are treated with Fermi-Dirac statistics and a carrier-dependent mobility ($\mu(n)$). [56] The results of the Semiconductor step is a spatial distribution of the carrier density that is modified by the presence of the metallic tip above the surface. We use the Drude model to calculate a spatially-dependent permittivity at a given frequency ω based on the solved concentration of electrons and holes for a given location: [55]

$$\varepsilon(\omega, \mathbf{r}) = \varepsilon_s \left(1 - \frac{\omega_p^2}{\omega(\omega + i\gamma)} \right) \quad (1)$$

where $\gamma = \frac{e}{\mu(n)m^*}$ is the carrier damping rate, and ω_p is the plasma frequency, which can be calculated for both electrons and holes:

$$\omega_p^2 = \frac{ne^2}{\varepsilon_s m^*} \quad (2)$$

In the equations above, $\varepsilon_s = 11.68\varepsilon_0$ is the static permittivity for Si, e is the charge of an electron, $m_e^* = 0.272m_0$ is the electron effective mass, $m_h^* = 0.367m_0$ is the hole effective mass, and n is the carrier density. [57] The resulting spatially-dependent permittivity is a sum of both electron and hole contributions and represents the change in the optical properties of doped Si due to tip-induced band-bending (TIBB) and can be used in a subsequent scattering simulation. Dopant density simulations are performed with a resolution of 10 points per dopant decade and the results are connected with a spline interpolation.

3. Results and discussion

3.1. Method validation & substrate effects

In order to validate our approach and demonstrate modeling of layered samples commonly encountered in s-SNOM, we use our model to predict near-field spectra for a thin layer of PMMA on glass and gold substrates (Fig. 1(A)). Spectral sweeps were performed with 10 cm^{-1} resolution and the resulting scattering signal was demodulated at 2Ω with $A_{\text{tap}} = 20$ nm in order to yield a near-field amplitude and phase (Fig. 1(B)). For an easy comparison between our FEM model and the multilayer-FDM (ML-FDM) and to highlight substrate effects, amplitude spectra were

normalized to a bulk PMMA response. For the PMMA/Au case, there is a strong increase in scattering amplitude relative to glass substrates (Fig. 1(B)). This is expected due to the large reflection coefficient of gold, which leads to an increased illumination of the tip apex and thus stronger overall scattering. [58] Our model also predicts a significantly stronger enhancement for the PMMA/Au case when compared to those predicted by the ML-FDM, suggesting the FEM is accounting for additional illumination mechanisms not considered in the ML-FDM, such as multiple reflections between the tip and sample. The predicted phase spectra show the strong, characteristic C=O absorption at $1730\text{--}1740\text{ cm}^{-1}$, as well as lower frequency modes corresponding to CH₂, C-C-O, and C-O-C stretches. For the PMMA/glass case, these lower frequency modes are obscured by the strong SiO₂ absorption from the substrate. We find good agreement between our results and experimental near-field phase spectra of PMMA on gold and glass. [59] Interestingly, there is increased phase contrast for the C=O mode on the glass substrate, which has also been observed in previous calculated spectra for the PMMA/glass case and related substrate near-field studies. [13,60] However, this higher phase contrast is difficult to observe in experiment, as the reduced scattering on glass substrates lowers the signal-to-noise. [60]

The large near-field phase predicted in the region from 1000 cm^{-1} to 1250 cm^{-1} for the PMMA/glass is due to the strong absorption by the Si-O vibration in glass in the range between $\approx 1000\text{--}1300\text{ cm}^{-1}$. [61] In the Supplementary Material (Figure S3), we show the calculated electric field distributions that confirm this strong absorption from the substrate and also reveal unique field distributions at these frequencies. At a frequency of 1000 cm^{-1} the electric field underneath the tip is dipolar, which mirrors the near-field on a gold surface at all frequencies studied (Figure S3B). As the frequency increases towards the SiO₂ resonance, we observe a large electric field at the PMMA/glass interface. The field reverses sign in the PMMA domain and resembles an antisymmetric surface mode. Surface phonon-polaritons (SPhPs) have been observed previously in other SiO₂ polymorphs. [61] The excitation of these polariton modes can lead to unique near-field phenomena. [37,62] While the amorphous structure of glass produces significant loss that likely inhibits the observation of propagating SPhPs, the strong response of the Si-O absorption at this frequency still gives rise to a non-dipolar distribution which contributes to the large near-field phase observed. This behavior is not observed for the strong C=O mode at a frequency of 1730 cm^{-1} , which highlights the differences in tip-interactions with molecular oscillators and solid state materials. The complex field distribution observed for the PMMA/glass substrates suggests that strong tip-sample coupling will have important ramifications for near-field experiments on polaritonic materials, which we explore in the next section.

3.2. Strong coupling effects

Having benchmarked our model for the layered PMMA case, we turn to a more interesting scenario involving strong tip-sample coupling with another well-known material, 4H-SiC. Here FEM models are expected to be more accurate than analytical models, as the electric field underneath the tip is no longer strictly dipolar due to resonance effects. [37,62] For these models, we investigate the spectral region from $840\text{--}960\text{ cm}^{-1}$, which is part of the highly-reflective Reststrahlen band in SiC located between the longitudinal optical (LO) and transverse optical (TO) phonon modes. [63] At these frequencies, light couples strongly to phonons in SiC and enables the excitation and propagation of SPhPs, collective mode excitations that, like surface plasmons, have been of intense interest in the literature. [21–23,42,64–68]

The interaction and coupling of SPhPs with the tip in s-SNOM can lead to large signal enhancement and a complex optical response. [9,37,64] To demonstrate this, we use our model to predict the change in scattering as the tip nears the surface at a frequency of 936 cm^{-1} . Figure S4 and Fig. 2 show the calculated electric field and approach curves for a Pt ellipse over a gold

and SiC surface respectively. For the gold case, the electric field between the tip and gold surface remains qualitatively the same, retaining its dipolar character at all distances. Furthermore, the calculated amplitude approach curve monotonically increases as the tip nears the surface, with no detectable change in phase beyond numerical noise (Figure S4B). However, as the distance between the tip and SiC is decreased, the electric field in the gap between the tip and sample becomes increasingly multipolar, with a minimum in the field immediately below the tip at a distance of $z = 1$ nm (Fig. 2(A)). This deviation from the typical dipolar behavior is observed even at moderate tip-sample distances ($z = 40$ nm).

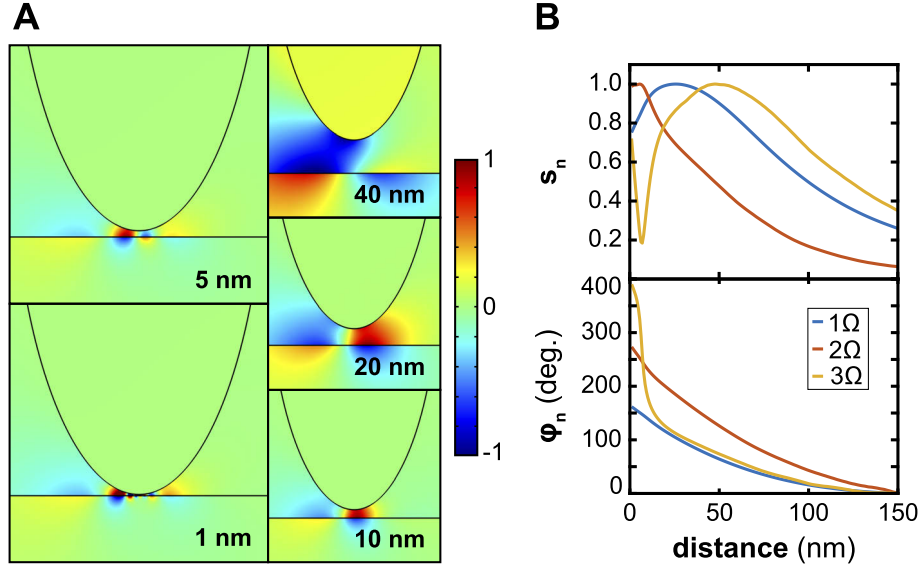


Fig. 2. Near-field distribution & approach curves for a Pt ellipse over 4H-SiC. (A) Vertical electric field distribution of an ellipse over 4H-SiC near its resonance frequency (936 cm^{-1}) and at different tip-sample separations. (B) Near-field amplitude (top panel) and phase (bottom panel) approach curves at the same frequency and for different demodulation orders ($n\Omega$, $A_{\text{tap}} = 35 \text{ nm}$).

This near-field behavior has also been observed in simulations of other resonant materials such as Au close to its plasmon resonance, [62,69] and arises from field back-action and the excitation of surface waves with large wavevectors. [37] This strong distance-dependent behavior has a significant effect on the approach curves typically collected in near-field experiments. In Fig. 2(B), we show the calculated near-field amplitude and phase approach curves for the first three demodulation orders. Unlike approach curves on non-resonant or weakly-resonant surfaces such as gold, [6] we observe non-monotonic behavior in the amplitude for the first three demodulation orders, which all differ slightly in their behavior. There is also a large increase in phase as the tip approaches the surface, which is further exaggerated due to the demodulation process. Along with the non-monotonic amplitude, the large near-field phase relative to the incident field helps explain the experimental observation of non-monotonic near-field intensity approach curves, [70] as the measured, demodulated intensity (I_n) is proportional to both the amplitude and the phase of the field underneath the tip:

$$I_n \approx S_n e^{i\phi_n} \quad (3)$$

As a complementary study, we investigate the spectral response of this strong coupling and its impact on the demodulation process. Figure 3(A) displays vertical electric field distributions

for gap modes at three different frequencies for a Pt ellipse 1 nm away from 4H-SiC. At lower frequencies (888 cm^{-1}), the electric field displays the typical dipolar mode associated with tip-sample coupling. As the frequency is increased closer to the resonance (928 cm^{-1}), the gap mode takes on increasingly multipolar character, similar to the 1 nm case in Fig. 2(A).

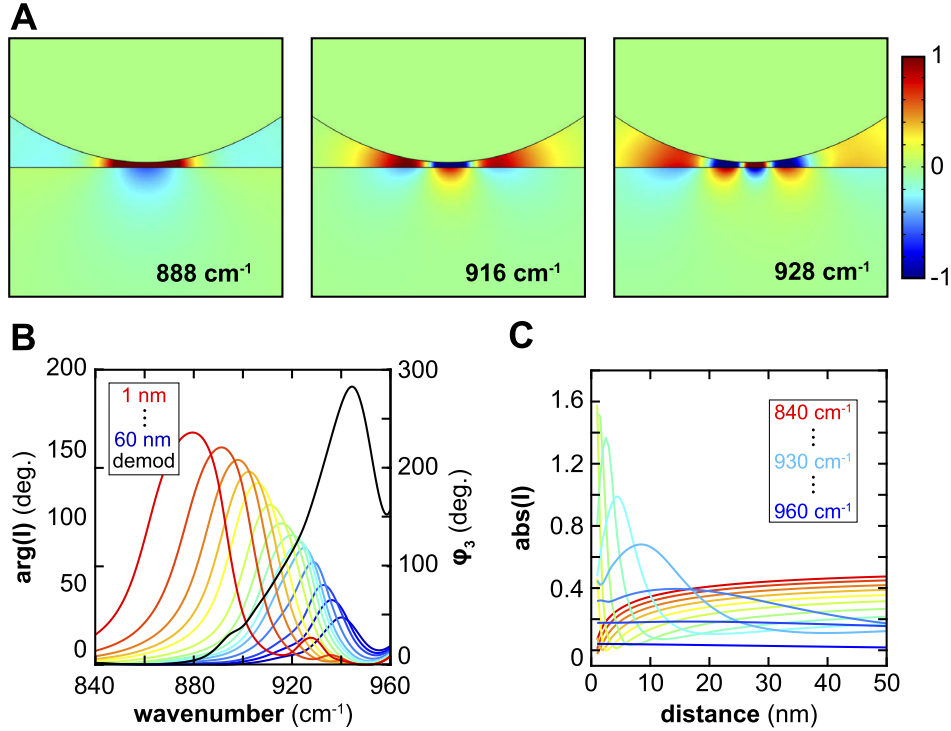


Fig. 3. Simulated near-field spectra for a Pt ellipse over 4H-SiC. (A) Vertical electric field distribution underneath an ellipse 1 nm away from 4H-SiC at three different frequencies. (B) Near-field phase spectra for 4H-SiC at various tip-sample separations (colored lines) and for the demodulated case (2Ω , $A_{\text{tap}} = 35\text{ nm}$). (C) Intensity approach curves for 4H-SiC demonstrating the non-monotonic behavior at frequencies ranging from 840 cm^{-1} to 960 cm^{-1} .

Near-field phase (Fig. 3(B)) and amplitude (Figure S5) spectra were calculated for various tip-sample distances (left-axis) and for a demodulated case (2Ω , $A_{\text{tap}} = 35\text{ nm}$). We observe a peak in the scattering amplitude near the SPhP resonance that redshifts by over 50 cm^{-1} as the tip approaches the surface. This red-shifting of vibrational modes in near-field experiments is a well-known effect. [71] However, the red-shift is typically small for weak coupling cases. [59] The significant red-shift at small tip-sample separations has been attributed to multiple reflections in studies using analytical models based on Green's functions, [72–74] although the influence of multipolar effects may also be present in our model. For demodulated spectra, this leads to a large degree of broadening, due to the large range of distances sampled during tapping mode. This broadening observed in our simulations is supported by near-field spectra we have collected on a 4H-SiC wafer (MTI Corp.) at various tapping amplitudes (see Figure S6). The broadening from demodulation can obscure other modes, such as the higher-order peak at a frequency of 916 cm^{-1} for the $z = 1\text{ nm}$ spectrum, highlighting the importance of electromagnetic modeling for strongly-resonant systems. The overall scattering peak, however, still occurs closer to the expected resonance at 943 cm^{-1} , where $\varepsilon \approx -1$. This is likely the result of two effects. Firstly, the majority of the red-shift occurs mostly at small tip-sample distances ($z < 10\text{ nm}$), which is only a

small portion of the overall tip motion for moderate tapping amplitudes. In other words, because the tip spends more time farther away from the sample, its response at larger distances contributes more to the demodulation process. Secondly, the intensity approach curves become increasingly strong and non-monotonic as the frequency approaches 930 cm^{-1} due to excitation of SPhPs (Fig. 3(C)). Thus, the shape of the approach curve is strongly dependent on the incident frequency due to the strong tip-sample interaction and this leads to a stronger signal at higher harmonics. This has ramifications for experiment, as the selection of which harmonics to monitor is often guided by the assumption that the signal reduces with increasing harmonic, which may not always be the case (Figure S7). In summary, strong interactions between the tip and sample can create large qualitative differences in near-field observables, and our results highlight the importance of collecting multiple harmonics in experiments involving strongly-resonant materials. The simulations performed here demonstrate a complex distance-dependent behavior which could have potential applications for modifying the near-field polarization or providing access to novel excitations by controlling the tip-sample separation. [75]

3.3. Tip electrostatic effects

s-SNOM has become more popular for semiconductor characterization in recent years, due to its ability to measure the electrical properties of samples with high-spatial resolution and without the need for contacts. For example, s-SNOM has been used for qualitative carrier profiling in doped semiconductors such as Si, [8,28,30,76] InP, [29] InAs, [77] GaAs, [78] and ZnO. [79,80] In these studies, the conductivity is measured optically through the free carrier absorption, which is sensitive to the number of free carriers present near the surface. The tips used in these s-SNOM experiments are typically coated with Pt or Au, metals chosen for their large polarizability and thus signal enhancement. However, tips such as these can introduce artifacts due to electrostatic forces, [81] and in semiconductors can significantly modify the local free carrier density, even for non-contact modes. [82] This tip-induced band-bending (TIBB) has been investigated for scanning tunneling microscopy (STM) and scanning microwave impedance microscopy (sMIM), [82,83] but there have been no studies to date of TIBB effects in IR s-SNOM. To accomplish this, we leverage the ability of FEM simulations to incorporate multiple types of physics, combining the Wave Optics and Semiconductor modules to simulate band-bending and its effect on the overall infrared near-field signal. This model uses the scattering simulations developed for the PMMA and SiC cases, but it is preceded by a Semiconductor step that solves for the expected carrier density beneath the tip. This carrier density is then combined with the Drude model to define a spatially dependent permittivity that can be used in the scattering simulations to account for TIBB effects in s-SNOM.

We first apply our model in the infrared spectral range, where the sensitivity of s-SNOM lies in the high-doping regime: $1 \times 10^{18}\text{ cm}^{-3} < N_D < 1 \times 10^{21}\text{ cm}^{-3}$. [28] In general, the s-SNOM response curve on doped Si has a dispersive lineshape, centered on $N_D \approx 3.5 \times 10^{19}\text{ cm}^{-3}$, due to a resonance in the polarizability of the coupled tip-sample system at this frequency (926 cm^{-1}). [34] The position of this resonance is the point where the real part of the n-Si permittivity is $\epsilon \approx -1$ and determines the sensitivity range of s-SNOM for a given excitation frequency. As shown in the Supplemental Information (Figure S8), we observe a very slight electron depletion from upward band-bending at the surface, leading to a minor shift in the near-field response (Figure S8B,C). The low impact of TIBB in this case is a result of the high carrier density already present in the Si, which can readily screen the tip's potential and prevent any large changes in the carrier density and electric field at the surface. Thus, TIBB effects are expected to be very small for mid-infrared measurements, where the sensitivity of s-SNOM lies at high-doping regimes.

However, TIBB effects can be significant for low to moderately-doped semiconductors, which correspond to an s-SNOM sensitivity in the THz region. [49,84] To demonstrate this, we repeat our simulations at longer wavelengths corresponding to THz frequencies (20 cm^{-1} or

$\lambda = 500 \mu\text{m}$). Figure 4(A) shows the electron distribution due to TIBB for a dopant density of $1 \times 10^{16} \text{ cm}^{-3}$. Unlike the infrared case, we observe a strong depletion of carriers at the surface, which significantly modifies the silicon refractive index (Fig. 4(B)). To explore this effect further, we repeated our simulations across multiple dopant densities ($1 \times 10^{15} \text{ cm}^{-3} < N_D < 1 \times 10^{19} \text{ cm}^{-3}$). As a control case, we also performed our simulations without including any TIBB effects (solid black line). Figure 4(C) and Figure S9 show the predicted dependence of the near-field (2Ω) phase and amplitude against doping concentration. Qualitatively, the results of the FEM simulations without TIBB align closely with predictions by the FDM (dashed black line). Similar to the PMMA/Au case discussed above, we again observe a stronger scattering behavior in our FEM simulations compared to the FDM, which we attribute to the inclusion of multiple reflections between the tip and sample that lead to a stronger illumination of the tip. When TIBB is included (solid red line) there is a significant change in the near-field response due to the modified optical properties of the surface. The large degree of electron depletion at the surface causes the overall free carrier absorption and conductivity to be lower and thus the near-field response is shifted towards higher dopant densities. We find that the relative error between an s-SNOM measurement with or without TIBB is greatest at the dopant density that s-SNOM is most sensitive to for this wavelength (Figure S10). The onset of depletion occurs while the tip is relatively far from contact, with a 10x reduction in the surface electron density occurring at a distance $z \approx 30 \text{ nm}$. As the tip approaches the surface, the magnitude of the depletion grows and extends further into the bulk, with depletion volumes that are comparable to those probed by s-SNOM, highlighting the importance of this phenomenon. Furthermore, instead of the single phase peak at $N_D \approx 3 \times 10^{16} \text{ cm}^{-3}$ observed for the FDM and non-TIBB cases, we observe two peaks at $N_D \approx 6 \times 10^{16} \text{ cm}^{-3}$ and $2 \times 10^{17} \text{ cm}^{-3}$. To investigate this effect further, we show the dependence of the near-field phase on both demodulation order and tapping amplitude in Fig. 4 and Figure S11 respectively. The prominence of both peaks increases at higher demodulation orders and lower tapping amplitudes, which suggests the behavior is caused by the response close to the surface. Based on this observation, we hypothesize this phase behavior is due to a complex interplay between the s-SNOM probing depth and the tip-induced spatial distribution of carriers at the surface. Our results illustrate the importance of considering TIBB in near-field studies on semiconductors.

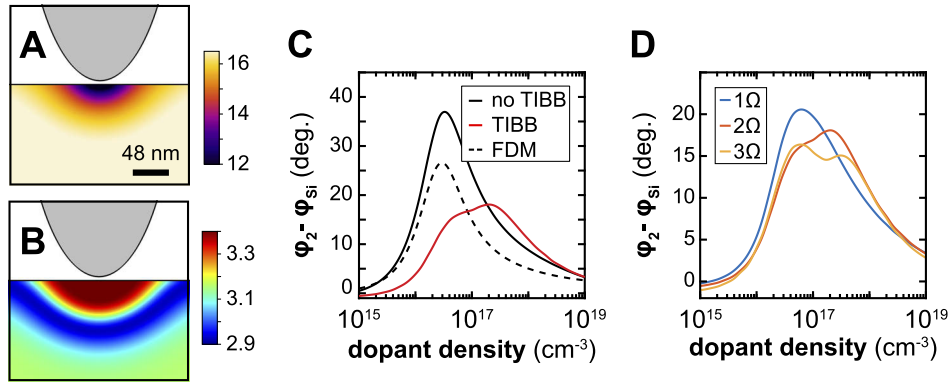


Fig. 4. Distribution of (A) electrons (cm^{-3} , log-scale) and (B) silicon refractive index illustrating the degree of band-bending from a Pt ellipse 5 nm away from the n-Si surface at a THz frequency (20 cm^{-1} and $N_D = 5 \times 10^{16} \text{ cm}^{-3}$). (C) Calculated near-field phase (2Ω , $A_{\text{tap}} = 20 \text{ nm}$) for a Pt ellipse over a n-type Si surface without (solid black) and with band-bending effects (solid red). The results from the finite-dipole model (dashed black) are included for comparison. (D) Dependence of the near-field phase in (C) on demodulation order ($n\Omega$).

As a demonstration of how spatial behavior and TIBB effects can impact measurements on semiconductor junctions or devices, we applied our TIBB simulations to a metal-semiconductor junction. The silicon is treated as n-type with native doping $N_D = 1 \times 10^{16} \text{ cm}^{-3}$ and the metal contact has a work function of $\phi = 4.7 \text{ V}$. Thus there is band-bending both near the junction and underneath the metallic tip, leading to a complex two-dimensional carrier distribution. For an incident wavelength of $\lambda = 600 \text{ nm}$, we simulate the undemodulated near-field signal ($\tilde{S}$) for a Pt tip $z = 2 \text{ nm}$ way from the surface as it scans across this junction with 50 nm steps (Fig. 5). While we observe a significant difference between the two cases for the near-field phase, we observe very little change in the scattering amplitude. One possible explanation for this is shown in Figure S6, which shows only a small change in the scattering amplitude at lower doping concentrations, amounting to a $\approx 5\%$ difference for the doping level considered here. On the other hand, the change in phase when TIBB effects are included is significant. For both scenarios, we see a small increase in amplitude and phase as the tip crosses the junction followed by a decrease as the tip moves further into the semiconductor. This can be explained by considering the fact that the scattering in a near-field experiment is not the result of the sample's response directly below the tip apex but by an averaged area comparable to the tip radius. Thus, the scattering from the tip is ultimately determined by an effective permittivity related to the probing area and probing depth of the tip. When the tip approaches the junction from the highly-conductive metal side, the effective permittivity starts to increase towards the tip-sample resonance condition, which leads to an increase in both scattered amplitude and phase. Interestingly, the near-field signal falls slowly as the tip continues away from the junction and its depletion region. This could be a result of the fact that our simulations are undemodulated and thus contain more background effects than would be observed in an experiment.

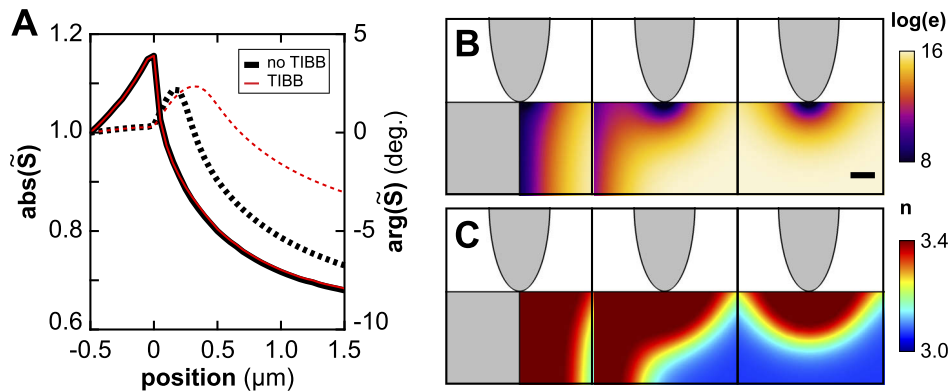


Fig. 5. (A) Predicted near-field amplitude (solid lines) and phase (dashed lines) across the junction for simulations with TIBB effects (thin red) and without TIBB effects (thick black). Distribution (B) of electrons and the resultant Si refractive index (C) at three tip positions ($x = 0, 0.3, 1.0 \text{ μm}$). Scale bar = 100 nm .

The simulations presented here demonstrate the importance of TIBB for interpretation of s-SNOM results on semiconducting samples or devices with moderately-doped regions. We note that in these studies we have not included any surface-specific effects, such as surface recombination or interface states, which would further modify the results. Those phenomena can be added to the model with the Semiconductor module for specific materials, as required. Although we only considered a planar case here, we anticipate tip-induced band bending to be even more significant for nanostructured geometries, especially for small dimensions ($< 50 \text{ nm}$) and/or low dopant densities. [82]

4. Conclusions

Here we have developed finite-element models based on COMSOL Multiphysics for a wide range of scenarios that are not easily accessible with traditional analytical models used in near-field microscopy. We first validate our approach on a layered PMMA sample and use this as an opportunity to discuss substrate effects and highlight differences between the FDM and FEM simulations. We then apply our model to SiC, whose complex, multipolar response, has important ramifications for the demodulation process and impacts measured approach curves and spectra. Finally, we incorporate electrostatic effects into our model to predict the effects of tip-induced band bending on s-SNOM measurements of semiconductors, which show a significant shift in the THz s-SNOM response for low doped semiconductors. While our TIBB simulations have only considered $V_a = 0$, an applied tip bias will modify the effect, and highlights the importance of TIBB for other scanning probe techniques. We have focused on these specific model examples, but we emphasize that FEM simulations can easily be scaled to any frequency provided appropriate experimental or theoretical values for the dielectric function of the material can be determined. Our model also demonstrates that FEM simulations are an appropriate choice for modeling strong tip-sample coupling, where multipolar fields are significant and the typical dipolar assumptions about the near-field are no longer valid. Furthermore, the use of FEM allows the incorporation of physical phenomena that are of significant interest in near-field experiments but not previously accessible, such as electrostatic, thermal, and strain effects. These models would represent a large advancement in understanding the s-SNOM response on more complex systems such as heterostructures, waveguides, and nanomaterials.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See [Supplement 1](#) for supporting content.

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